

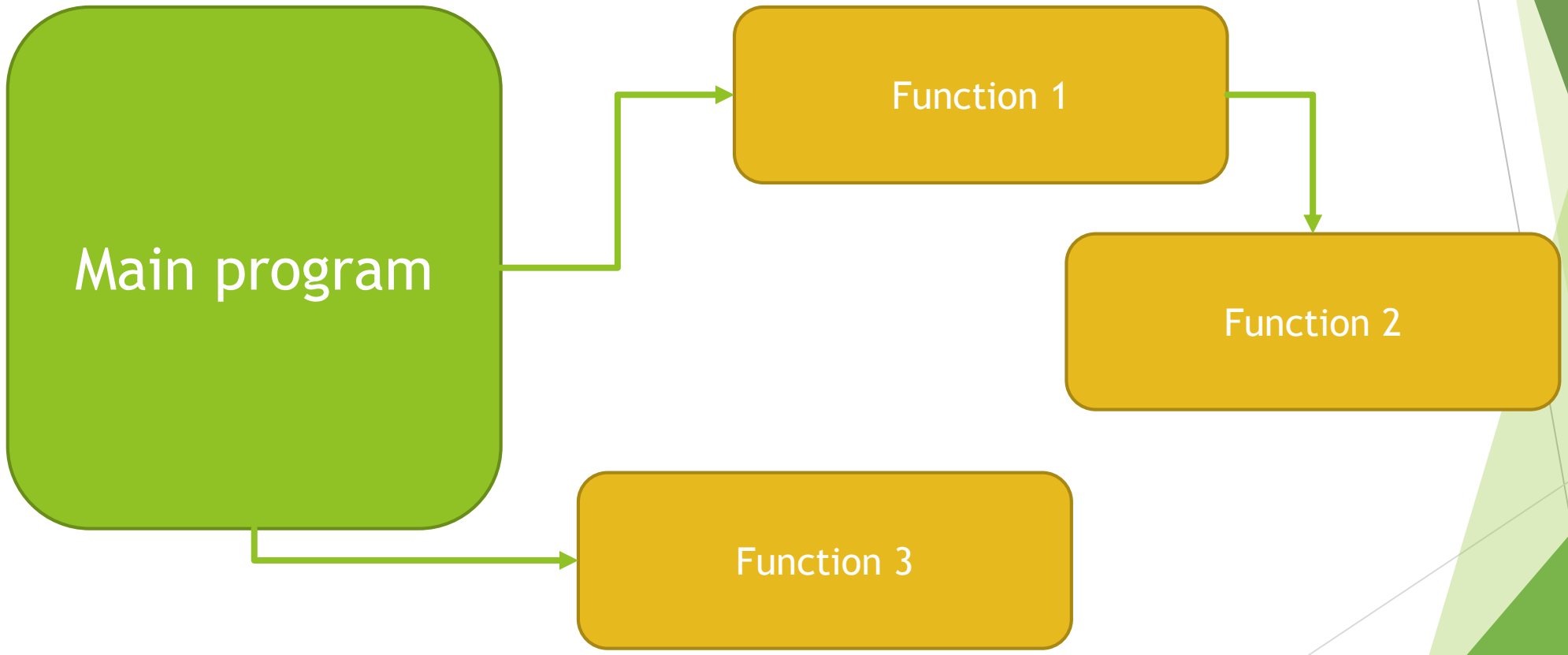
Functions

Modular programming

What is a Function?

- A small code block
- Separate from the main process
- Main process can call the block when needed
  - This can be called multiple times
- They are capable of taking values and returning values
- Very helpful to address complex programming scenarios
  - Helps to simplify coding work

Visual Representation



Some Rules and Norms

- Starts with the word **def**
  - Then function name and parentheses
- Function name cannot starts with a number
  - Numbers may appear middle or at the end
- No spaces are allowed in function names
- Functions may accept values (as arguments) if required
- Functions may return values (as results) if required

Types of Functions

1. Simple functions
2. Functions accept parameters
3. Functions return values
4. Functions accept parameters and return values

Your First Simple Function

```
# Simple fucntion
def MyFun():
    "This line is for nothing"
    print ("I'm from a function")
    return

# ------------------------ Starting main body ---

print ("Main program body")
# Calling the fucntion
MyFun()
print ("End of program")
```

Your First Simple Function [Explain]

```
# Simple fucntion
def MyFun():
    "This line is for nothing"
    print ("I'm from a function")
    return

# ------------------------ Starting main body ---

print ("Main program body")
# Calling the fucntion
MyFun()
print ("End of program")
```

Function Accepting Arguments

```
# Simple fucntion
def MyFunArg(value):
    "This line is for nothing"
    print ("Multiply by 2 is:",value * 2)
    return

# ------------------------ Starting main body ---

print ("Main program body")
# Calling the fucntion
MyFunArg(15)
print ("End of program")
```

Function Accepting Arguments [Explain]

```
# Simple fucntion
def MyFunArg(value):
    "This line is for nothing"
    print ("Multiply by 2 is:",value * 2)
    return

# ------------------------ Starting main body ---

print ("Main program body")
# Calling the fucntion
MyFunArg(15)
print ("End of program")
```

Function Returning Values

```
# Simple fucntion
def MyFunReturn():
    "This line is for nothing"
    MySpell = "Accio Lumos Maxima"
    return MySpell

# ------------------------ Starting main body ---

print ("Main program body")
# Calling the fucntion
print ("Harry Potter said,", MyFunReturn())
print ("End of program")
```

Function Returning Values [Explain]

```
# Simple fucntion
def MyFunReturn():
    "This line is for nothing"
    MySpell = "Accio Lumos Maxima"
    return MySpell

# ------------------------ Starting main body ---

print ("Main program body")
# Calling the fucntion
print ("Harry Potter said,", MyFunReturn())
print ("End of program")
```

Function Accept and Return Values

```
# Simple fucntion
def MyFunAcceptReturn(value1, value2):
    "This line is for nothing"
    answer = value1 + value2
    return answer

# ------------------------ Starting main body ---

print ("Main program body")
# Calling the fucntion
print ("Addition of 2 and 2 is", MyFunAcceptReturn(2,2))
print ("End of program")
```

Function Accept and Return Values [Explain]

```
# Simple fucntion
def MyFunAcceptReturn(value1, value2):
    "This line is for nothing"
    answer = value1 + value2
    return answer

# ------------------------- Starting main body ---

print ("Main program body")
# Calling the fucntion
print ("Addition of 2 and 2 is", MyFunAcceptReturn(2,2))
print ("End of program")
```